



PART 1

Chapter 2: Software Process

2.1. Giving reasons for your answer based on the type of system being developed, suggest the most appropriate generic software process model that might be used as a basis for managing the development of the following system:

- A system to control anti-lock braking in a car

Ans: Safety-oriented system requires heavy analysis and design -> Waterfall model.

- A virtual reality system to support software maintenance

Ans: Rapidly changing requirements -> Incremental development model

- A university accounting system that replaces an existing system

Ans: Well known requirements + integration with other systems -> Reuse-oriented SE model

2.2. Explain why incremental development is the most effective approach for developing business software systems. Why is this model less appropriate for real-time systems engineering?

Ans:

2.4. Suggest why it is important to make a distinction between developing the user requirements and developing system requirements in the requirements engineering process.

Ans: (nên viết thành đoạn từ ý chính)

User requirements:

- Stem from users
- Users must understand the requirements
- Should be written in natural language and allow flexibility
- Comes first

System requirements:

- Stem from development
- Adapt to the current available tools and technologies the development team has access to
- Should be detailed
- Comes after

They are fundamentally different from each other

2.5. Describe the main activities in the software design process and the outputs of these activities. Using a diagram, show possible relationships between the outputs of these activities.

PART 2

Sao nguyên đồng này not english bruv

CHƯƠNG 1:

1. What is a software process, software process activities?

A software process is a set of related activities that leads to the production of a software product.

4 Activities:

- Software specification
- Software design and implementation

- Software validation
- Software evolution

CHƯƠNG 2:

1. Phân biệt các quá trình – tính chất (có thể hỏi TN)
2. V model có thi

- i. Waterfall model: traditional approach
 1. Sequential
 2. Making sure one stage completed before doing the next
 3. Suitable for projects having stable requirements: it is eliminating the waste of projects.
 4. Stages: requirements definition, design & analysis, implementation, testing, deployment, maintenance & operation.
 - ii. RUP: traditional approach
 1. Iterative: doing things repeatedly, incrementally
 2. Phases: Inception, Elaboration, Construction, and Transition
 3. Iterations = cycle
 - a. One iteration is like one cycle of waterfall
 4. Detailed process to do things
 5. Detailed roles of team members
 6. Many documents to be written
 7. Use-case driven development
 - iii. Agile methods
 1. Four value propositions of Agile methods
 2. Scrum
 - a. Sprint, time-boxed iteration
 - b. Backlog (product backlog, sprint backlog, impediment backlog)
 - c. Roles
 - d. Activities
 - i. Daily scrum
 - ii. Sprint review
 - iii. Sprint planning
 - iv. Release planning
 3. XP
 - a. Philosophy: take down good practices into extreme
 - b. Key practices
 - i. Pair programming
 - ii. Small release
 - iii. Coding standard
 - iv.
-

CHƯƠNG 3:

1. Vai trò PM là gì, các hoạt động mà ng quản lí dự án cần làm gì. xử lí tình huống: nếu ng muốn rời khỏi team, KH thay đổi ý định, xung đột trong nhóm. 90% cho thi đề xuất lên kế hoạch cho dự án đc đưa (cô nói dị)

The principal responsibility of software project managers is to manage the project so that the software is delivered on time and within the planned budget for the project. They supervise the work of software engineers and monitor how well the software development is progressing.

2. Success criteria ? <- ?

CHƯƠNG 4:

1. Phân biệt non-functional & functional requirements cho ví dụ

Functional requirements are statements of services the system should provide, how the system should react to particular inputs, and how the system should behave in particular situations (or should not to)

Non-functional requirements are constraints on the services or functions offered by the system.

2. Mqh giữa non-functional & functional requirements với kiến trúc hệ thống là gì? Cho 3 ví dụ (liên quan tới bài 6)
3. 3 essential non functional requirements for cái gì đó (vd ứng dụng phần mềm như coursera) explain why each has been priority

CHƯƠNG 5:

1. Cho mô tả vẽ bảng use case (gồm các mục: usecaseID, use case Name, brief description, actors, basic flow, alternative flows, special requirements, pre-conditions, post-conditions, extension points) à tự nhớ bảng, chứ đề ko nói

CHƯƠNG 6:

1. Học mấy cái architecture pattern (hỏi TN), khi nào dùng nó
2. Mqh giữa non-functional requirements vs architecture (cho 3 vd ngoài)

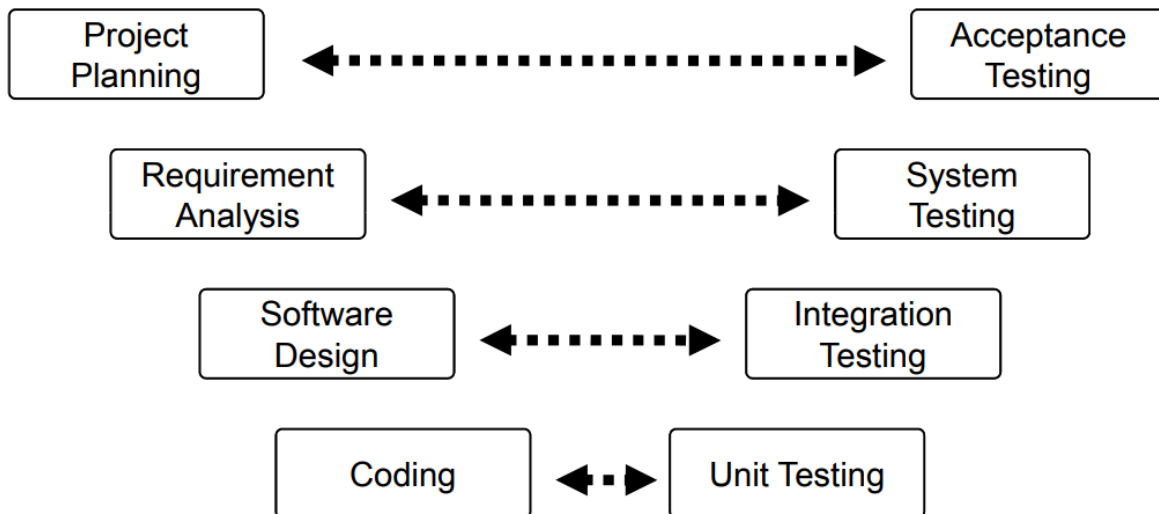
CHƯƠNG 8:

1. UI design principles
2. Đọc mô tả r thiết kế màn hình trên thiết bị gì đó (1 hoặc 2 màn hình), giải thích tại sao lại vẽ như vậy (trong CW8) có vẽ. Phương pháp là:
 - a. Xem coi 1 chức năng có thể đc chia ra bao nhiêu màn hình phù hợp

- b. Bao nhiêu lần thao tác (click chuột,...), tổng hợp các thao tác đó thành min nhất là chọn màn hình phù hợp

CHƯƠNG 9:

1. Thi có vẽ 4 test case cho 1 funct (gồm các mục: testcaseID, testcase Name, description, relate usecase, input data, expected output, test steps)
2. Inspections and testing + vd
3. Mô hình test (slide 14)
4. Nêu quy trình test (slide 15), tại sao pha đó quan trọng trong quá trình (có thể là hỏi pha validation quan trọng trong quá trình phát triển phần mềm ntn)

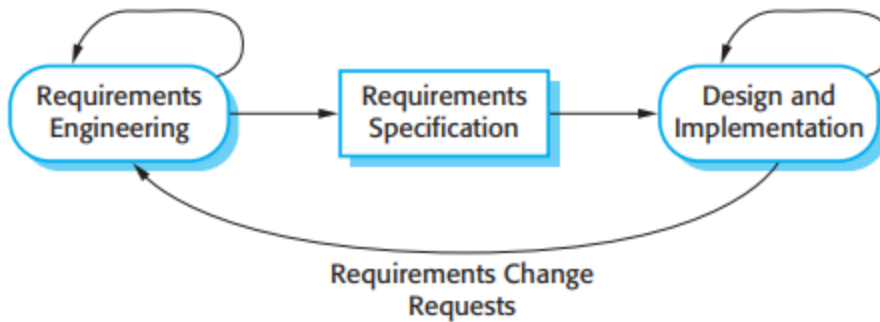


5. Test plan

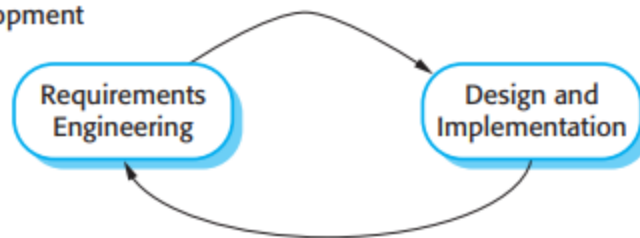
CHƯƠNG 10:

1. So sánh plan-driven & agile. Trình bày pp, cho vd, khi nào dùng

Plan-Based Development



Agile Development



2. Đặc điểm : agile methods, manifesto, 5 nguyên tắc của agile và vận dụng

Manifesto:

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan

Principle	Description
Customer involvement	Customers should be closely involved throughout the development process. Their role is provide and prioritize new system requirements and to evaluate the iterations of the system.
Incremental delivery	The software is developed in increments with the customer specifying the requirements to be included in each increment.
People not process	The skills of the development team should be recognized and exploited. Team members should be left to develop their own ways of working without prescriptive processes.
Embrace change	Expect the system requirements to change and so design the system to accommodate these changes.
Maintain simplicity	Focus on simplicity in both the software being developed and in the development process. Wherever possible, actively work to eliminate complexity from the system.

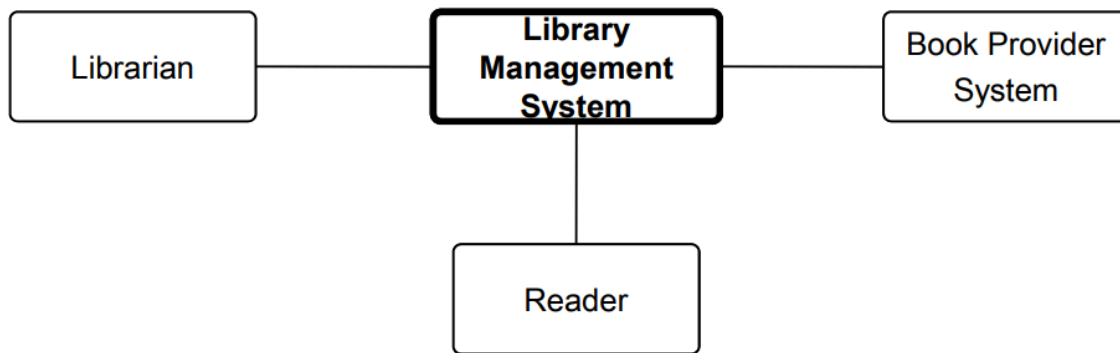
3. Đặc điểm của scrum <- ?

PART 4

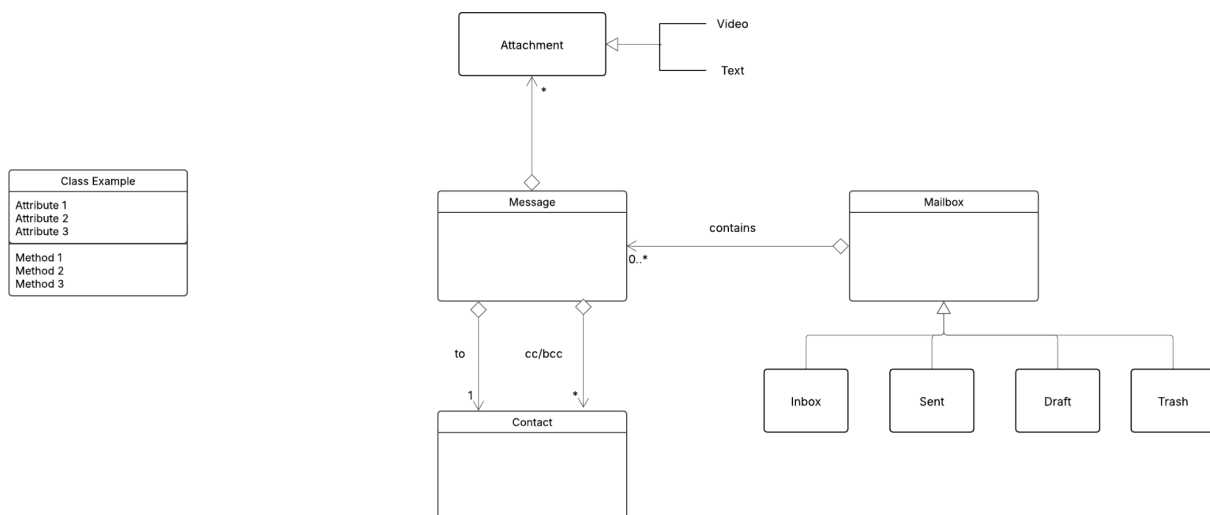
Types of Diagrams:

Context Diagram:

- Main Box of the system with all the other boxes as stakeholders



Class Diagram:



Sequence Diagram (per Use Case):

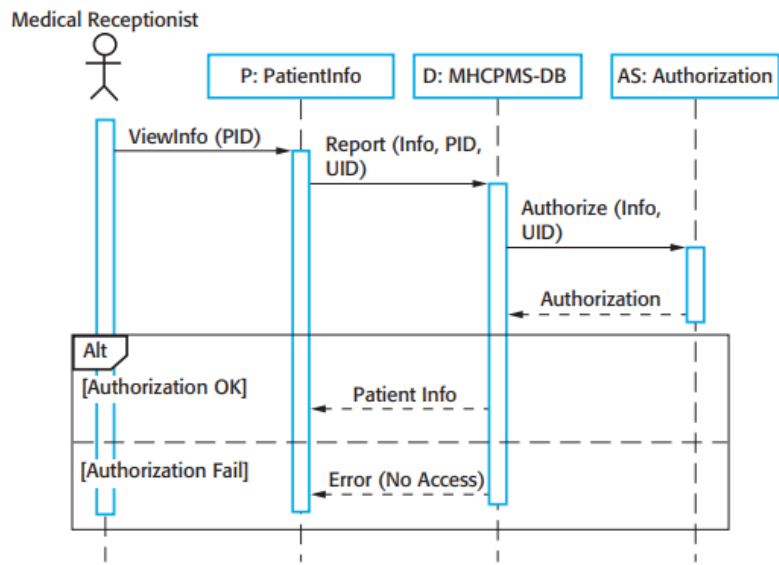
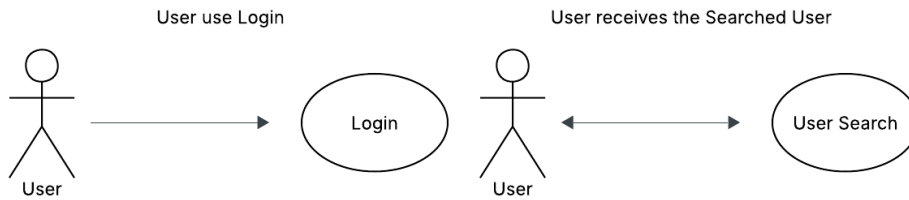
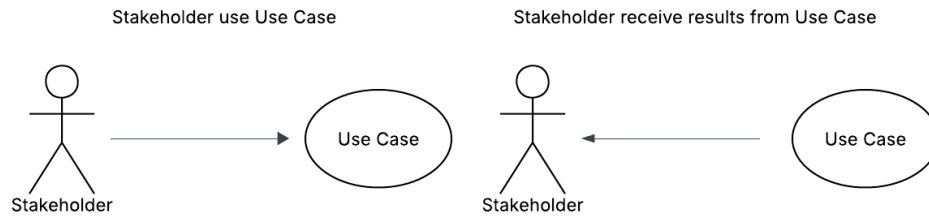
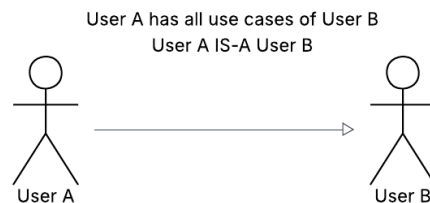
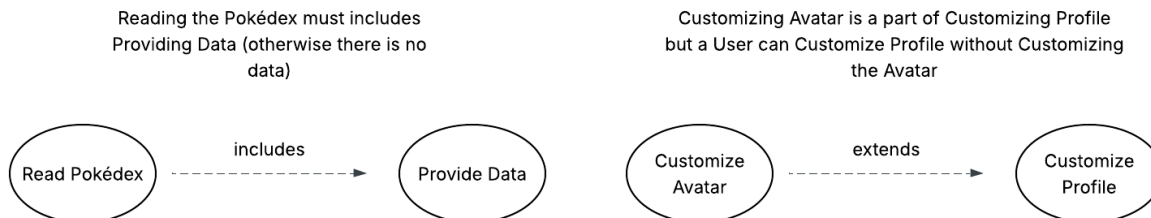
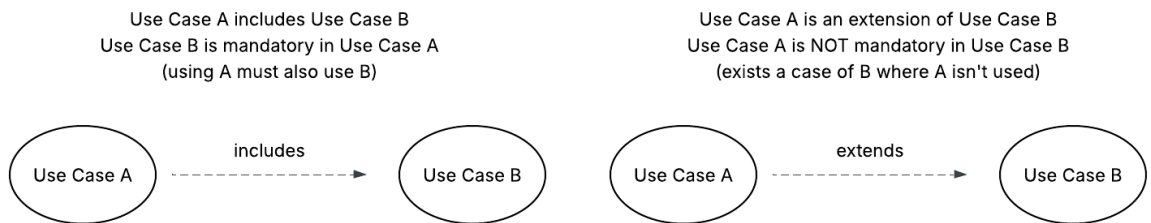


Figure 5.6 Sequence diagram for View patient information

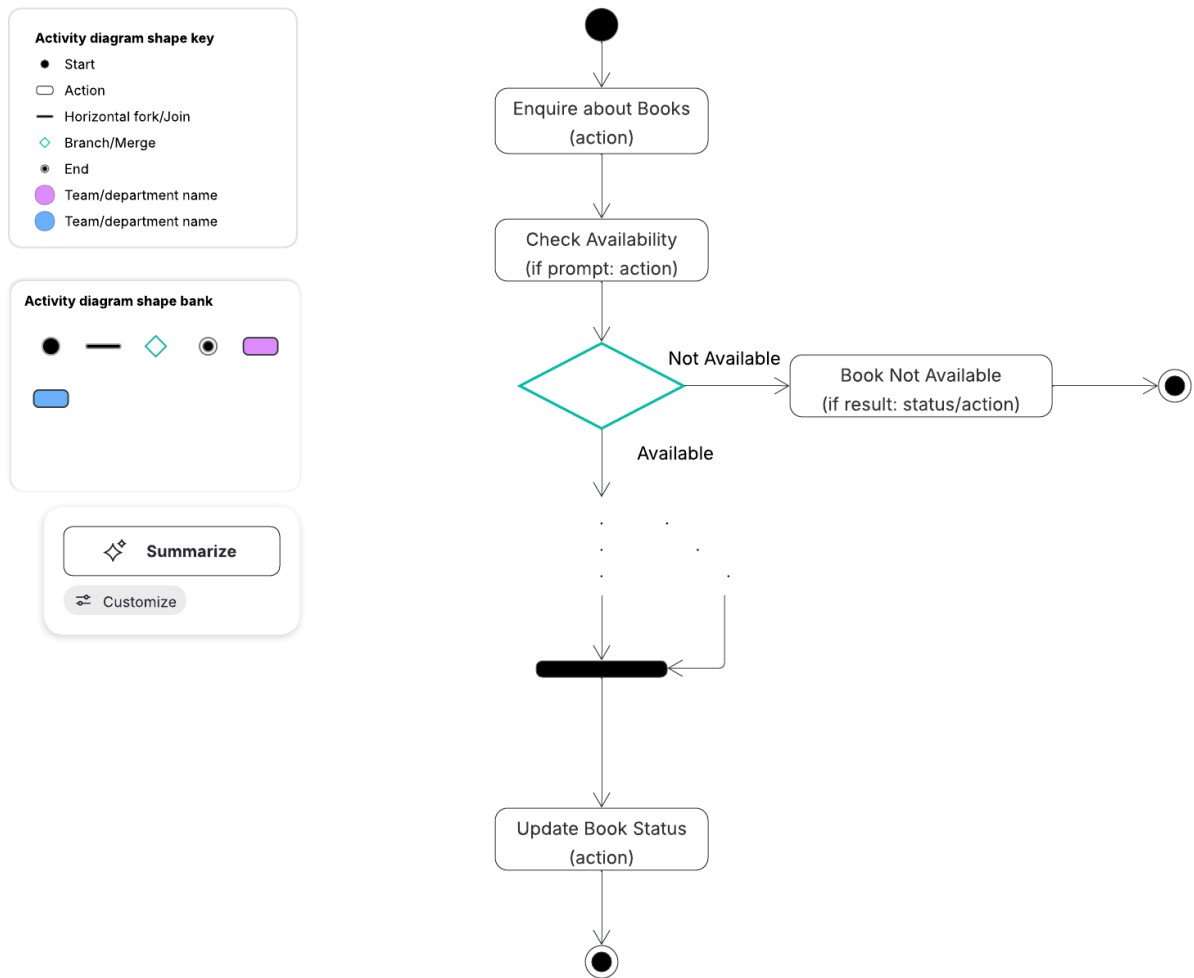
Use Case Diagram:



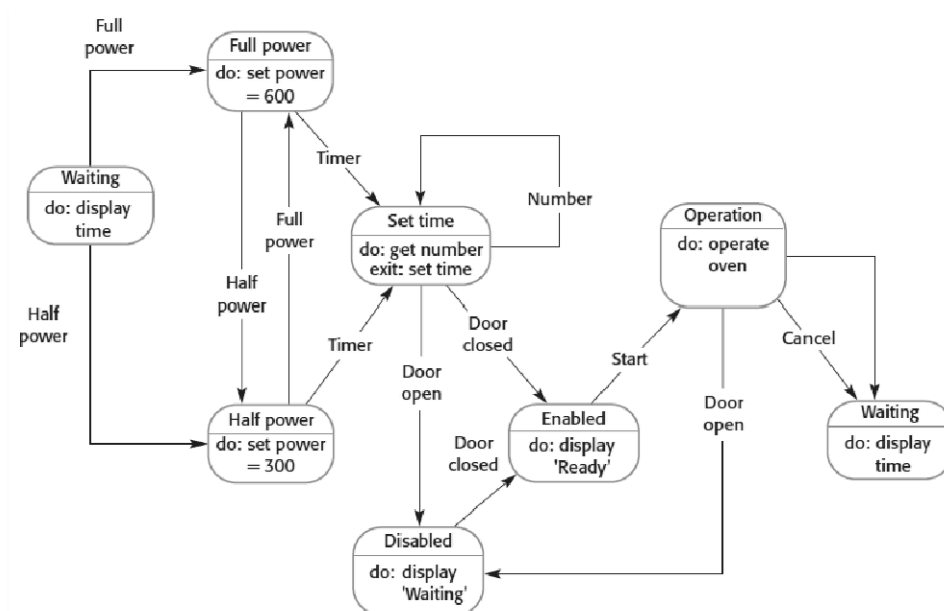
Note: A, B must be distinct enough from each other



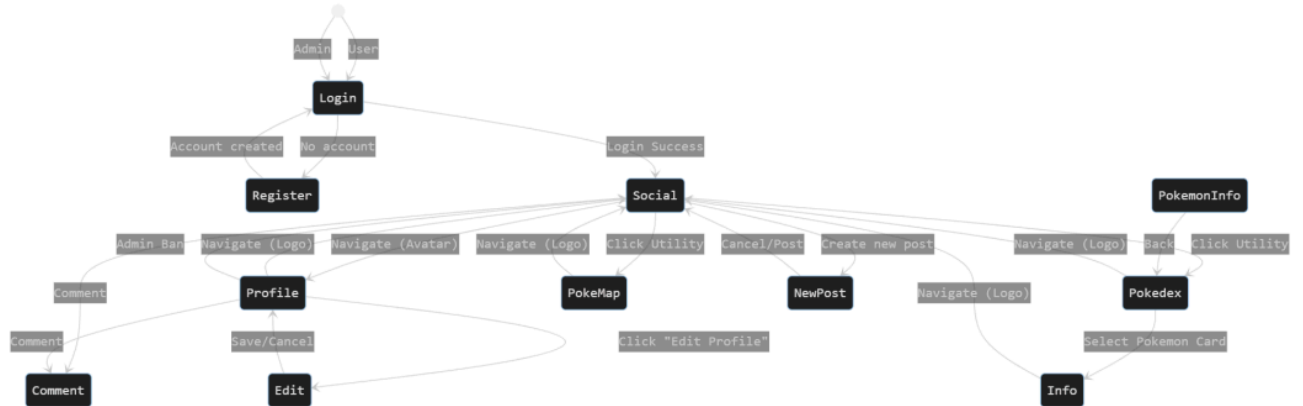
Activity Diagram:



State-Chart Diagram of an Oven:

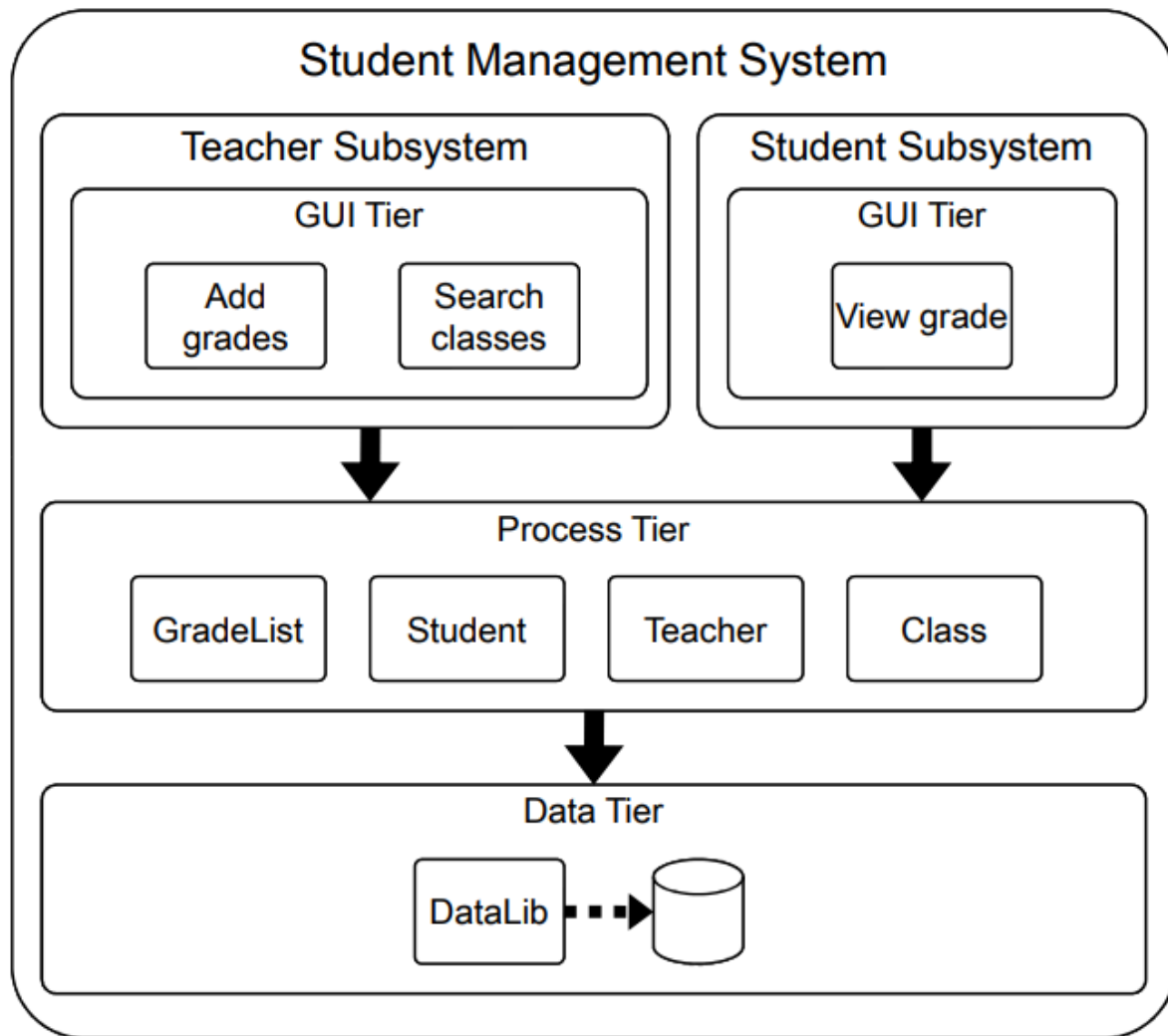


Screen Diagram:

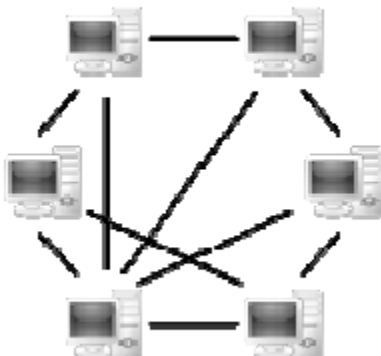


Architecture:

- Monolithic (1-Tier) Model
- Client-Server (2-Tiers) Model
- 3-Tiers Model:



- Peer-to-Peer Model (overly simplified):



Types of Tables:

Class Specification for User Class:

3.3.1 Class User

Attributes

Seq	Property	Modifier	Constraint	Description
1	UserID	Private	Primary Key	A unique identifier for the User.
2	Username	Private	Mandatory	The user's displayed name.
3	Email	Private	Mandatory, Validated Email	The user's email address.
4	DoB	Private	Mandatory	The user's date of birth.

Methods

Seq	Property	Modifier	Constraint	Description
1	follow(userID)	Public	Requires valid userID	Allows the current user to follow another specified user.
2	view(mapID)	Public	Requires valid mapID	Records that the user has selected to view the map.
3	makeComment(postID, commentID, content)	Public	Requires valid postID and commentID	Creates a new comment on a specified post with the given content.
4	makePost(postID, ...)	Public	Requires valid postID	Creates a new post with the given details as parameters.

Use Case Specification cho Login Use Case:

<i>Use case ID</i>	U002
<i>Use Case</i>	Login
<i>Brief Description</i>	This feature is accounted for when a user logs into the website.
<i>Actor</i>	User(Player), Account System
<i>Pre-Condition</i>	The user must be at the login form.
<i>Result</i>	The user is logged in to the system and gains access to the main webpage.
<i>Main Scenario</i>	1. User enters their registered username and password. 2. User presses the login button. 3. System verifies the credentials against the database. 4. System authenticates the user and loads their account data. 5. System updates the interface to the main page.
<i>Alternative Scenarios</i>	3a. System detects that the username does not exist. 4a. System returns to the login form while displaying an error to the User about invalid credentials. 3b. System detects that the password is incorrect. 4b. System returns to the login form while displaying an error to the User about invalid credentials.
<i>Non-Functional Constraints</i>	The credential verification process must be completed within 3 seconds. Error messages must be displayed for a minimum of 5 seconds.

Use Case Analysis cho Login:

Scene	Actor	Screenplay
Login (tên use case)	Reader/Librarian (tên những stakeholder sử dụng use case)	Login successfully scenario (tên các trường hợp trong use case) -System: request username and password. -User: provide username and password. -User: submit login. -System: validate username and password. Introduction to Software Engineering - Nguyễn Minh Huy 24 -System: validate username and password. -System:

		redirect user to main screen.
		Login failed scenario At step 4 of success scenario: -System: validate username and password. -System: report error and request user re-login.

Test Case Specification: (nên thêm ID)

Test Case	Nhập sách thất bại
Related Use Case	Add new book
Context	Login as Librarian
Input Data	Book title: Chiến tranh và Hòa bình Author: (empty) Publisher: (empty) Episode: 2 ...
Expected Output	Display error message "Invalid book information!". The book is not stored in database.
Test Steps	1. Login: (username "nmhuy", password "12345") 2. Select "Add new book" on menu 3. Enter Input Data. 4. Click submit button.